Predicting Reasoning Robustness from Hidden-State Stability

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Code, data lists and submission bundles: `https://github.com/yupengtang/aimo-interp-challenge-2026`. Codabench participant `ytang`.

1 Summary

The nominated method never generates text. For each problem it shows the evaluated model the original statement and six rewrites that cannot change the answer, runs one forward pass per view, and measures how far the model's own hidden state and next-token distribution move. A logistic regression over those movements, calibrated separately for each checkpoint, gives the verdict. Every number it uses is read from inside the model being judged.

Under the per-problem evaluation the organizers introduced on 15 October, it scores 13/21 on the Main track and 9/14 on the Small track. Removing the white-box component leaves only its constant fallback, which scores 11/21 and 7/14 on the same sets.

Method	Main (21 cases)	Small (14 cases)	Signal
Representation stability (nominated)	13/21 = 0.62	9/14 = 0.64	evaluated model's hidden states
Same method without the white-box component	11/21 = 0.52	7/14 = 0.50	none (constant)
Problem-level judge (black-box comparison)	13/21 = 0.62	8/14 = 0.57	gpt-oss-120b's answers
The judge read from gpt-oss hidden states (Section 7, white-box)	11/21 = 0.52	7/14 = 0.50	gpt-oss-120b's hidden states

The comparison method, a judge that asks gpt-oss-120b how settled it is on the problem, reaches 0.84 accuracy on the official training labels and falls to 0.62 on validation. The organizers later explained why: validation was built so that every problem has both robust and non-robust configurations, and the test set follows the same construction. A verdict that depends only on the problem is then close to chance by design. The stability signal is computed per model, from that model's own internals, so it can disagree across configurations of the same problem. That is the property the evaluation rewards, and the main reason we nominate it.

Section 5 adds what we found about which configuration of a problem breaks: in the labelled data the configuration's own accuracy on the original problem separates the two in a U shape. We could not measure it well enough within the time limit to use it.

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2 The setting

`are_robust(model_id, problems, reasoning_effort)` runs offline on one GPU. The task and its labelling are described in the challenge proposal [Stefánik et al., 2026]. Since 15 October the harness calls it once per case with a one-problem list, under a limit of 70 minutes for the run. A case is robust when no perturbation lowers the model's accuracy by more than 0.10, and not robust when some perturbation lowers it by 0.25 or more; cases in between are left out.

Two properties of the data shape everything below. In training, labels follow the problem: `aimo-interp/train-main-v2` has 71 labelled rows over 36 problem statements, fifteen problems are labelled for two or more configurations, and fourteen of those have the same label for every configuration, so predicting a row from its siblings is right 50 times in 51. Validation and test are built the other way. Every validation problem has at least one robust and at least one non-robust configuration (organizers, by email, October 2026), and the public validation sets are balanced overall as well: always-robust scores 10/21 and 7/14, never-robust 11/21 and 7/14.

3 The nominated method

Views. Each problem gets six rewrites in three families: two transpose letters inside long prose words, two insert a neutral sentence that carries no quantity, and two add a neutral wrapper before or after the statement. LaTeX spans and Asymptote blocks are never edited. Rewrites are seeded by problem id, so they are identical across runs.

White-box features. One forward pass per view over the chat-formatted prompt of the evaluated checkpoint. Between the original and each rewrite we take the cosine similarity of the last prompt token's hidden state at four depths, and the KL divergence of the next-token distribution. These are aggregated as mean and minimum within each family, plus the prompt length, giving 31 features. Depths are named by quartile of the layer stack, so one coefficient vector fits checkpoints with different layer counts. Every feature compares a model with itself, so the features are dimensionless and comparable across checkpoints in a way raw hidden states are not.

Training. L2-regularised logistic regression with $C = 0.003$ on 154 DeepSeek-R1-0528-Qwen3-8B cases from the public MATH labels [Hendrycks et al., 2021] (38 robust), with the olympiad problems held out. C was chosen by cross-validation grouped by problem text over five seeds; in-domain grouped CV AUC is 0.693.

Calibration. For each checkpoint, features are standardised with statistics from 226 problems whose labels we did not use: 216 public olympiad problems from AIME and AMC collections, and 10 MATH problems whose only labels belong to a non-competition model. The threshold is the median score on that pool, so the detector calls about half the cases robust, which matches the balanced construction of validation and test.

Runtime. Views run one at a time, on purpose: a padded batch moves hidden states enough to flip one verdict in six for two checkpoints, and the artifact was fitted on unpadded features. On the competition worker under per-problem calls every case finished in 5 to 16 seconds (21 cases in 189 s on Main, 14 in 81 s on Small), with no timeouts and no case answered by the fallback. If time ever ran short, rewrite families are dropped in order, and any failure returns the constant fallback, so a case cannot become invalid.

4 Ablation: the white-box component

The method has exactly one non-white-box path: when the hidden-state features cannot be computed it returns the constant "not robust". Removing the white-box component therefore turns it into the never-robust baseline, which we scored on the same validation sets.

Track	With hidden-state features	Without (constant fallback)	Difference
Main	13/21 = 0.62	11/21 = 0.52	+2 cases
Small	9/14 = 0.64	7/14 = 0.50	+2 cases

The run logs confirm that no case fell back to the constant, so every verdict in the first column came from the model's internals. Two cases per track is a small margin on sets this size (one case is worth 4.8 points on Main and 7.1 on Small), and we do not claim more than that the white-box features carry the method above the baseline on both tracks.

On the training labels the picture is weaker. On the 34 labelled rows of the four small checkpoints in `train-main-v2` the method scores accuracy 0.44 and AUC 0.42 [0.18, 0.66]. We read this together with Section 2: train labels follow the problem, so a per-model signal has little to gain there, while validation rewards exactly the within-problem separation a per-model signal can provide. Both samples are small, so we offer this as an interpretation only.

5 What separates configurations within a problem

Robustness is a small maximum drop in accuracy under perturbation. A configuration that never solves the original problem has nothing to lose, so it is robust by construction. One that always solves it is robust unless a perturbation breaks it. One that solves it only part of the time can lose accuracy from sampling alone, because the drop is a maximum over several noisy estimates. That predicts a U shape in the configuration's own accuracy, and the labelled data shows one.

On the public sample (`aimo-interp-challenge-sample-full`: 10 problems, 8 configurations, the competition labelling rule, expert-written perturbation types excluded), the robust share by the configuration's own accuracy is:

Own accuracy on the original problem	Rows	Robust
below 0.05	20	100%
0.05 to 0.95	23	43 to 50%
0.95 or more	28	82%

"Robust if own accuracy is below 0.05 or at least 0.95" reaches 0.78 accuracy and 0.75 balanced accuracy there. With the expert "no solution" perturbation included the top group drops to 36% robust: a configuration that always answers loses accuracy when it keeps answering a problem that has no answer.

The same pattern appears on `train-main-v2`. With the majority answer of all gpt-oss samples as reference, configurations matching it in at least 95% of their samples are robust in 21 of 22 rows; those between 50 and 95% in half the rows. The small checkpoints almost never finish within the 8192 tokens we could afford, so their near-zero accuracy mostly reflects truncation.

We tried two black-box ways of using this, and both did worse than the judge. Inverting the judge for gpt-oss configurations scored 11/21 on Main, against 13/21. Sampling each gpt-oss configuration at its own effort (8 samples, 4096 or 6144 tokens, robust if at least 7 match the reference) scored 0.75 accuracy on train against 0.83 for the judge, because unfinished high-effort samples counted as mismatches, and it would also need about 2600 s for 18 problems on an H200.

This is the strongest argument we have for a white-box route: the quantity that separates configurations is a property of the evaluated model on that problem, and reading it from a few forward passes is far cheaper than estimating it by sampling.

6 The black-box comparison: a problem-level judge

For every problem not seen before in the run, gpt-oss-120b [OpenAI, 2025] gets the organizers' solve prompt at low reasoning effort and is sampled sixteen times (temperature 0.6, top-p 0.95, at most

2560 new tokens). The problem is called robust if at least 13 samples give the same answer. The verdict is reused for every configuration asked about that problem.

On the 71 labelled training rows the share of the modal answer has AUC 0.90 [0.78, 0.97], and the rule resamples to 0.84 accuracy and 0.85 balanced accuracy. On validation it scored 13/21 and 9/14 under the old protocol and 13/21 and 8/14 under per-problem calls. When the judge is settled on a problem, the case is robust in 43 of 45 train rows but in 6 of 10 validation cases on each track. With two configurations of a problem carrying opposite labels, any one-verdict-per-problem rule is right on exactly one of them; with more configurations it can only match each problem's majority. Train, where labels follow the problem, cannot show this ceiling.

Our first estimate of the judge was too optimistic. A first run with four samples per problem made "all four agree" score 0.87 balanced accuracy. A second independent draw scored 0.76, and resampling put the rule at 0.68 with a standard deviation of 0.07. The first draw was about two and a half standard deviations lucky, and we had chosen the threshold on it. Every judge estimate above is therefore a resampled mean.

7 The judge, read from hidden states

The judge works, but it costs about 50 s of sampling per problem, and it is black-box. We asked whether gpt-oss already knows how settled it will be before it writes anything. For 720 unlabelled olympiad problems from Omni-MATH [Gao et al., 2024] (stratified by its difficulty rating, every problem that also appears in `train-main-v2` removed) we ran the judge exactly as in the bundle (8 samples, low effort, 3072 tokens) and recorded the share of the modal answer. We also kept gpt-oss's hidden states for the same chat prompt, without generating: the last prompt token and the mean over the prompt, at layers 9, 18, 27 and 36. A ridge probe per feature predicts the modal share; the verdict averages the eight probes, each standardised by its own cross-validated scores, and the threshold calls half of the pool robust. All of these choices were fixed before looking at the official labels.

Signal	Cost per problem	AUC on the 71 labelled training rows
Judge, modal-answer share (8 samples)	~50 s of generation	0.76
Probe on gpt-oss hidden states (8-probe average)	one forward pass, ~0.2 s	0.70
Representation stability (Section 3)	7 forward passes of the evaluated model	0.42
Text only: fewer $ signs	none	0.71
Text only: shorter statement	none	0.65

On the training rows the probe is close to the judge and far above the stability method, but it is not separable from a count of $ signs: 36 problems cannot tell them apart. The pool can. Across 720 problems the cross-validated probe correlates 0.66 with the judge's modal share, while statement length and $ count correlate 0.32 and 0.27. Most of what the probe reads is therefore not surface form. Removing the white-box component again leaves only the constant fallback. Run as the competition harness runs it, the bundle loads gpt-oss once (565 s cold on our cluster) and then answers 36 problems in about 7 s, with no case on the fallback.

The probe inherits the judge's limitation: it gives one verdict per problem, so on per-problem-balanced data it can only match each problem's majority (Section 6). On validation it scored 11/21 on Main and 7/14 on Small, exactly the never-robust baseline (scored on the organizers' temporary evaluation queue during the Codabench outage of late October). That is the ceiling of Section 6 made concrete: a signal read from the problem alone, however well it ranks problems, cannot separate the configurations that validation was built to separate. We report it as a cheap white-box copy of the judge; it is not one of our nominated submissions. Two of the 36 training problems got a different verdict in the bundle run than in the offline features; gpt-oss is a mixture-of-experts model, and tiny numerical differences between GPUs can change the experts a token is routed to.

8 What did not work

Approach	Evidence	Why
Last-token hidden-state probe trained on MATH	MATH AUC 0.74; 0.41 to 0.65 on olympiad problems	In-domain CV ranks encoders almost opposite to transfer
Answer agreement under cheap rewrites, same model	AUC 0.385 at 8192 tokens	No signal once truncation is handled
Label-free domain standardisation	Moves AUC by at most 0.03	Fixes calibration, not ranking
Training only on longer MATH problems	CV falls from 0.693 to 0.56	The signal lived in the short problems
Self-labels from answer-format perturbations	Agreement 0.31, majority baseline 0.76	These perturbations barely break the model
Per-checkpoint priors	Probes within one case of the baseline	Section 2
The evaluated model's own uncertainty over a greedy 2048-token trace	Best AUC 0.71 [0.56, 0.85]; no gain added to the judge	No small checkpoint finishes in 2048 tokens
Judge's reaction to contradictions it writes itself	AUC 0.74 to 0.79 alone on train; no gain on train or validation	Redundant once agreement is measured well
Judge inverted for gpt-oss; own-effort sampling	11/21; 0.75 on train	Section 5

On MATH, "robust" mostly means "hard": longer problems, longer traces and failing to finish all predict a robust label there. The robust rate by prompt-length tertile is 0.05, 0.35 and 0.33, and prompt length alone reaches grouped CV AUC 0.713. On `train-main-v2` it is the easy problems that are robust. Anything fitted on the MATH sample should be checked for this sign flip before it is trusted on competition problems.

Truncation can also fake a consistency signal. With a 4096-token budget, 45% of DeepSeek samples had no answer, against 22% at 8192. Counting an unfinished sample as a disagreement turns consistency into a measure of trace length.

9 Limitations

The public validation sets have 21 and 14 cases. The white-box margin over the constant baseline is two cases per track, and the stability method and the judge tie on Main. The classifier was trained on one model's MATH labels, and its training-set accuracy on competition checkpoints is poor (Section 4), so its behaviour on the private problems is uncertain. The median threshold assumes a balanced test set, which the organizers' construction suggests but did not state for test. The first submission of this method scored 13/21 and then 12/21 on the same public set, a nondeterminism we did not trace; the per-problem run reported here is a single run.

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AI Use Statement

I used Claude Code (Anthropic) as an assistant for ideation, coding, experiment analysis and drafting the text. I did not use generative AI tools to set the research direction or to make the final choice of which submissions to nominate and report; I made those decisions myself. I have reviewed all AI-assisted work and take full responsibility for the final content of this paper.

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A Reproducing the methods

Environment. torch 2.12.1 with CUDA 13.2, transformers 5.13.0, scikit-learn 1.8.0 and NumPy, matching the competition image. Inference needs only NumPy and joblib beyond torch and transformers, because the artifact stores plain arrays rather than fitted estimators.

Labelled training data. `aimo-interp/augmented-sample-math-agg`, `-agg-filtered` and `-full-filtered`. A case is robust when `n_detrimental_permutations` is zero. After holding out the olympiad problems this gives 154 DeepSeek cases.

Unlabelled calibration data. `AI-MO/aimo-validation-aime`, `AI-MO/aimo-validation-amc`, `opencompass/AIME2025` and `Maxwell-Jia/AIME_2024`, deduplicated, with any problem that appears in the official data removed (216 problems), plus 10 problems from `hendrycks-math-test-reference-100` whose labels exist only for Llama-3.1-8B; those labels are not read.

Steps. `code/perturb_stability.py` computes the 31 features for every checkpoint. `code/train_stability_model.py --out stability_crossmodel.joblib` fits the classifier and the per-checkpoint calibration. The bundle in `submission_stability/` is `solution.py`, `scoring.py`, `perturbations.py`, `budget.py` and `stability_crossmodel.joblib`; the Small-track zip adds an empty `small.txt`. `code/trainv2_bundle_eval.py` scores it on the official training rows.

Comparison method and Section 5. The judge is `submission/solution.py` (a single file). `code/nosol_experiment.py`, `code/judge_rules.py` and `code/trainv2_bundle_test.py` reproduce Section 6; `code/own_acc_rule.py` reproduces Section 5.

Section 7 (the judge read from hidden states). The pool is the 720 Omni-MATH problems listed in `data/pool_omni720_ids.txt` (`KbsdJames/Omni-MATH`, test split, by row index). `code/distill/judge_pool.py` writes the judge targets (with `code/distill/judge_v3.py`, the 8-sample judge) and the hidden states for the pool and for the 36 training problems; `code/distill/fit_probe.py` produces the comparison table; `code/distill/export_probe.py <results> 0.5 probe.npz` fits the eight probes and the threshold. The bundle in `submission_whitebox_v2/` is `solution.py` and `probe.npz`; the Small-track zip adds an empty `small.txt`. Job scripts are `jobs/aimo_distill.sbatch`, `jobs/fit_probe.sbatch` and `jobs/aimo_wb2_test.sbatch`.

Determinism. Rewrites are seeded by problem id and inference does no sampling, so the stability verdicts depend only on the model's forward passes.